

Simon Oddershede Gregersen

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DBLP <https://dblp.org/pid/263/1059>

Employment

2026 – **CISPA Helmholtz Center for Information Security**. Tenure-track faculty.
2024 – 2025 **New York University**. Postdoctoral fellow. Advised by Joseph Tassarotti.
2023 – 2024 **Aarhus University**. Postdoctoral researcher. Advised by Lars Birkedal.
2017 **Aarhus University**. Center for Advanced Software Analysis. Student programmer.
2015 – 2017 **CCI Europe A/S (now Stibo DX)**. Student software developer.

Education

2017 – 2023 **PhD in Computer Science**. Aarhus University.
Advised by Amin Timay and Lars Birkedal.
Thesis: Higher-Order Separation Logic for Distributed Systems and Security.
2017 – 2020 **MSc in Computer Science**. Aarhus University.
2014 – 2017 **BSc in Computer Science**. Aarhus University.

Publications

- [14] Johannes Hostert, Zichen Zhang, Puming Liu, Simon Oddershede Gregersen, Ralf Jung, and Joseph Tassarotti. “Completeness of Iris-Based Program Logics”. In: *Proc. ACM Program. Lang.* 10.ICFP (2026). DOI: [10.1145/3828682](https://doi.org/10.1145/3828682).
- [13] Philipp G. Haselwarter, Alejandro Aguirre, Simon Oddershede Gregersen, Kwing Hei Li, Joseph Tassarotti, and Lars Birkedal. “Modular Verification of Differential Privacy in Probabilistic Higher-Order Separation Logic”. In: *Proc. ACM Program. Lang.* 10.PLDI (2026). DOI: [10.1145/3808311](https://doi.org/10.1145/3808311).
- [12] Simon Oddershede Gregersen, Chaitanya Agarwal, and Joseph Tassarotti. “Logical Relations for Formally Verified Authenticated Data Structures”. In: *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security, CCS 2025, Taipei, Taiwan, October 13-17, 2025*. 2025, pp. 1394–1408. DOI: [10.1145/3719027.3744801](https://doi.org/10.1145/3719027.3744801).
- [11] Kwing Hei Li, Alejandro Aguirre, Simon Oddershede Gregersen, Philipp G. Haselwarter, Joseph Tassarotti, and Lars Birkedal. “Modular Reasoning about Error Bounds for Concurrent Probabilistic Programs”. In: *Proc. ACM Program. Lang.* 9.ICFP (2025). DOI: [10.1145/3747514](https://doi.org/10.1145/3747514).
- [10] Philipp G. Haselwarter, Kwing Hei Li, Alejandro Aguirre, Simon Oddershede Gregersen, Joseph Tassarotti, and Lars Birkedal. “Approximate Relational Reasoning for Higher-Order Probabilistic Programs”. In: *Proc. ACM Program. Lang.* 9.POPL (2025). DOI: [10.1145/3704877](https://doi.org/10.1145/3704877).
- [9] Philipp G. Haselwarter, Kwing Hei Li, Markus de Medeiros, Simon Oddershede Gregersen, Alejandro Aguirre, Joseph Tassarotti, and Lars Birkedal. “Tachis: Higher-Order Separation Logic with Credits for Expected Costs”. In: *Proc. ACM Program. Lang.* 8.OOPSLA2 (2024). DOI: [10.1145/3689753](https://doi.org/10.1145/3689753).

- [8] Simon Oddershede Gregersen, Alejandro Aguirre, Philipp G. Haselwarter, Joseph Tassarotti, and Lars Birkedal. “Almost-Sure Termination by Guarded Refinement”. In: *Proc. ACM Program. Lang.* 8.ICFP (2024). DOI: [10.1145/3674632](https://doi.org/10.1145/3674632).
- [7] Alejandro Aguirre, Philipp G. Haselwarter, Markus de Medeiros, Kwing Hei Li, Simon Oddershede Gregersen, Joseph Tassarotti, and Lars Birkedal. “Error Credits: Resourceful Reasoning about Error Bounds for Higher-Order Probabilistic Programs”. In: *Proc. ACM Program. Lang.* 8.ICFP (2024). DOI: [10.1145/3674635](https://doi.org/10.1145/3674635).
- [6] Amin Timany, Simon Oddershede Gregersen, Léo Stefanescu, Jonas Kastberg Hinrichsen, Léon Gondelman, Abel Nieto, and Lars Birkedal. “Trillium: Higher-Order Concurrent and Distributed Separation Logic for Intensional Refinement”. In: *Proc. ACM Program. Lang.* 8.POPL (2024). DOI: [10.1145/3632851](https://doi.org/10.1145/3632851).
- [5] Simon Oddershede Gregersen, Alejandro Aguirre, Philipp G. Haselwarter, Joseph Tassarotti, and Lars Birkedal. “Asynchronous Probabilistic Couplings in Higher-Order Separation Logic”. In: *Proc. ACM Program. Lang.* 8.POPL (2024). DOI: [10.1145/3632868](https://doi.org/10.1145/3632868).
- [4] Simon Oddershede Gregersen, Johan Bay, Amin Timany, and Lars Birkedal. “Mechanized Logical Relations for Termination-Insensitive Noninterference”. In: *Proc. ACM Program. Lang.* 5.POPL (2021). DOI: [10.1145/3434291](https://doi.org/10.1145/3434291).
- [3] Léon Gondelman, Simon Oddershede Gregersen, Abel Nieto, Amin Timany, and Lars Birkedal. “Distributed Causal Memory: Modular Specification and Verification in Higher-Order Distributed Separation Logic”. In: *Proc. ACM Program. Lang.* 5.POPL (2021). DOI: [10.1145/3434323](https://doi.org/10.1145/3434323).
- [2] Morten Krogh-Jespersen, Amin Timany, Marit Edna Ohlenbusch, Simon Oddershede Gregersen, and Lars Birkedal. “Aneris: A Mechanized Logic for Modular Reasoning about Distributed Systems”. In: *Programming Languages and Systems - 29th European Symposium on Programming, ESOP 2020, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2020, Dublin, Ireland, April 25-30, 2020, Proceedings*. 2020. DOI: [10.1007/978-3-030-44914-8_13](https://doi.org/10.1007/978-3-030-44914-8_13).
- [1] Simon Oddershede Gregersen, Søren Eller Thomsen, and Aslan Askarov. “A Dependently Typed Library for Static Information-Flow Control in Idris”. In: *Principles of Security and Trust - 8th International Conference, Held as Part of the European Joint Conferences on Theory and Practice of Software, ETAPS 2019, Prague, Czech Republic, April 6-11, 2019, Proceedings*. 2019. DOI: [10.1007/978-3-030-17138-4_3](https://doi.org/10.1007/978-3-030-17138-4_3).

Awards

- Distinguished Paper Award at [ICFP 2024](#).
- Carlsberg Foundation Internationalization Fellowship ([CF23-0791](#)). DKK 1,020,000 ($\approx$ USD 150,000).

Advising

PhD students

- Aleksey Veresov (2026 –).

Interns

- Yanees Dobberstein (2026 May – July).

MSc Thesis Committees

- Peter Gastaur (2026). Saarland University.
Thesis: Compiling Distributed Algorithms in Pseudocode into Extended Threshold Automata.

Academic Service

Organizing

- 2027 Dagstuhl Seminar (no. 202606056). Foundations for Security of Cryptographic Protocols.
2026 Artifact Evaluation Co-Chair. **Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)** .

Program Committees

- 2027 Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA).
Certified Programs and Proofs (CPP).
2026 **Recent Advances in Concurrency and Logic (RADICAL)**.
SPLASH Onward!
Foundations of Computer Security (FCS).
Programming Language Design and Implementation (PLDI).
2025 **Recent Advances in Concurrency and Logic (RADICAL)**.

External Reviewing

- Journals **JFP** (2025), **TCS** (2024), **TOSEM** (2023)
Conferences **LICS 2025**, **LICS 2024**, **CCS 2022**, **POPL 2022**, **OOPSLA 2022**, **ICFP 2022**, **CSF 2022**, **CSF 2021**, **ESOP 2020**

Selected Talks

- 2026 June *Probabilistic Reasoning in Higher-Order Separation Logic*.
Invited, CISP/Inria Workshop.
2026 Feb *Building Extensible Program Logics through Effect Handlers*.
Invited, MPI-SWS Foundations of Programming group seminar.
2025 Dec *Building Extensible Program Logics through Effect Handlers*.
Invited, PLS Seminar, Aarhus.
2025 Oct *Logical Relations for Formally Verified Authenticated Data Structures*.
New England Systems Verification Day.
2025 Jun *Logical Relations for Formally Verified Authenticated Data Structures*.
The Fifth Iris Workshop.
2025 May *Logical Relations for Formally Verified Authenticated Data Structures*.
New Jersey Programming Languages and Systems Seminar.
2025 Apr *Logical Relations for Formally Verified Authenticated Data Structures*.
Invited, VU Amsterdam PLSec reading group.
2024 Apr *Trillium: Intensional Refinement in Higher-Order Separation Logic*.
New England Systems Verification Day.
2023 Jul *Asynchronous Probabilistic Couplings in Higher-Order Separation Logic*.
Invited, Bristol Programming Languages Research group seminar.
2023 Jul *Asynchronous Probabilistic Couplings in Higher-Order Separation Logic*.
Workshop on Verification of Probabilistic Programs (VeriProP).
2022 May *Trillium: History-Sensitive Refinement in Separation Logic*.
The Second Iris Workshop.
2020 Nov *Mechanized Logical Relations for Termination-Insensitive Noninterference*.
Invited, Chalmers ProgLog/Security seminar.